STA301 - Quiz No.03

Date: 08 January 2026

Q.1 Which of the following is correct property for joint probability distribution of X and Y:

A. None of the above

B. {f(Y,X)} = 1

C. {f(X,Y)} = 1

D. Both of the above

Q.2 Given below is the probability distribution of random variable X. What is the value of "k"

x	0	1	2	3	4
p(x)	0.70	k	0.10	0.07	0.03

B. 0.10

C. 0.01

D. 0.03

Q.3 Position of the normal curve changes along the horizontal axis due to:

A. Mean

B. Mode

C. Standard deviation

D. Median

Q.4 The parameters of binomial distribution are:

A. n,p,q

B. p,q

C. n,p

D. n,(n-1)

Q.5 The mean deviation of the normal distribution is approximately:

A. 7/8 of the S.D

B. 3/4 of the S.D

C. 1/2 of the S.D

D. 4/5 of the S.D

Q.6 Which one of the statement is true?

A. E (S2) = s2/n

B. E (S2) ≠ s2

C. E (s2) = 62

D. E (s2) = 62/n

Q.7 The marginal p.d.f. of _____ variables is obtained by integrating the other variable from the joint p.d.f.

A. Continuous

B. Qualitative

C. Binomial

D. Discrete

Q.8 In repeating sampling, which of the following possess desirable property of being Unbiased?

A. Sample proportion

B. Sample median

C. All of above

D. Sample Mean

Q.9 Which of the following statements is correct?

A. A point estimate is an unbiased estimator if its standard deviation is the same as the actual value of the population standard deviation

B. All of the above statements are correct

C. A point estimate is an estimate of the range of a population parameter

D. A point estimate is a single value estimate of the value of a population parameter

---

Q.10 Let X be a random variable with binomial distribution, that is (X=0,1,..., n). The expected value E[X] is

A. np

B. p

C. Xnp

D. np(1-p)
